# Bohao Peng

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## EDUCATION

**Harbin Institute of Technology, Shenzhen** 2018 – Present
Bachelor of Engineering - Computer Science and Technology
GPA: 93.11/100, 3.97/4; Bonus Points for Contests: 5/5
**Rank: 1/201**

## HONORS & AWARDS

- **National Scholarship** 2019
  - ✧ Top **1%**.
- **ACM – ICPC, International Collegiate Programming Contest, reginal: Gold Metal** 2020
  **ACM – ICPC, International Collegiate Programming Contest, reginal: Gold Metal** 2019
  **ACM – ICPC, International Collegiate Programming Contest, reginal: Silver Metal** 2018
  - ✧ Team leader.
  - ✧ Contribute first in the competition.
- **CCPC, Chinese Collegiate Programming Contest: Silver Metal** 2019
- **The Title of First Ten Top College Students in Harbin Institute of Technology** 2020
  - ✧ The highest honor for students.
  - ✧ The only student in the department of Computer Science and Technology.
- **Chunhui Innovation Achievement Scholarship** 2021
  - ✧ Top **0.1%**.
  - ✧ Reward for innovative invention in medical assistance systems.
- **The First-class Academic Scholarship** 2019&2020
- **Li Guang Scholarship** 2020
- **Outstanding Student in Harbin Institute of Technology** 2019&2020

## RESEARCH EXPERIENCE

**Sperm Morphology Analysis Based on Deep Learning and Shape Analysis** 2019 – Present
**Team leader, 15k+ lines of code**

- National Invention Patent: *Jingyong Su, Bohao Peng, Kezhou li. A Method and Device for Sperm Morphology Analysis. Patent Number: 202110083620.6.*
  - ✧ Develop a system to help doctors diagnose, which can automatically locate the sperms and analyze their morphology.
  - ✧ Complete the system architecture and hardware driver design, algorithm development with Pytorch and interactive interface development with PyQt5.
- Paper: *Bohao Peng, Kezhou Li, Dongzhi Jiang, An Automatic Sperm Analysis Method based on Semantic Segmentation and Shape Analysis.*
  - ✧ Design a novel two-stage framework with res-net and unet++. After object detection and sperm quality classification, semantic segmentation for contour detection and shape analysis

are used to deal with the problems due to imbalance of positive and negative samples in the dataset.

**Online Judge System for the Guangdong-Hong Kong-Macao Greater Bay Area Informatics Competition** 2020 – 2021
**Leader of live version, 3k+ lines of code**

- The online judge system is used to test programs in programming contests. The system can compile and execute the competitor's code, then compare the output with standard output and return the result.
- Develop functions iteratively for live competitions, such as communication between competitors and competition judges, message notification, etc.

**3D Subtomogram Alignment and Averaging** 2021 – Present

- Cryo-electron tomography (cryo-ET) allows the 3D visualization of a small volume of vitreous biological sample with nanometer resolution.
- Geometric unsupervised matching model finds the geometric correspondence between two images, and spatial transform network is used for 3D subtomogram alignment and averaging.

## TEACHING EXPERIENCE

**Instructor of ACM – ICPC Training Team** 2018 – Present

- Hold *China Collegiate Programming Contest (Final), 2018* as a student representative.
- Initiate and hold the first *Programming Contest of Harbin Institute of Technology (Shenzhen)* as the main person in charge.
- Teach courses includes:
  - ✧ Basic algorithm.
  - ✧ Advanced data structure.
  - ✧ Graph theory.
  - ✧ Discrete mathematics.

## SKILLS & COURSES

**Computer Skills:**

- ✧ C, C++, Python, Kotlin, Javascript, Verilog, etc.
- ✧ Linux, Git, Qt, Spider, Nodejs, MySQL, Pytorch, etc.

**Curriculum Highlights:**

- ✧ Mathematics: Set Theory and Graph Theory, Probability Theory, Mathematical Logic, Computational Methods
- ✧ Computer Theories: Data Structures, Digital Logic Design, Design and Analysis of Algorithms, Principles of Computer Organization, Computer Networks, Operating System

哈爾濱工業大學(深圳)

HARBIN INSTITUTE OF TECHNOLOGY, SHENZHEN

# 学生成绩单

## Student's Academic Transcript

**Name:**Peng Bohao

**Student ID:**180110110

**Entrance Time:**August,2018

**Program:**Undergraduate

**School:**School of Computer Science and Technology

**Major:**Computer Science and Technology

| Term | Course | Hours | Credit | Score |
|---|---|---|---|---|
| 18-19/Fall | Advanced Mathematics A | 80 | 5 | 90 |
| | Foundations of Computer Science | 32 | 2 | 91 |
| | English Listening and Speaking | 32 | 2 | 83 |
| | Linear Algebra and Analytic Geometry | 64 | 4 | 94 |
| | Ideological and Moral Self-cultivation & Fundamentals of Law | 48 | 3 | 95 |
| | Advanced Computer Language Programming | 48 | 3 | 95 |
| | Introduction to Cloud Computing | 16 | 1 | 92 |
| | Military Training and Theories | 3周 | 3 | 91 |
| | Physical Education A | 32 | 1 | 88 |
| | Set Theory and Graph Theory | 48 | 3 | 91 |
| | Entrepreneurship 101: Who is your customer? | 16 | 1 | 86 |
| | Wisdom of Business Ethics and Decision Making Between East and West | 28 | 2 | 93 |
| | Classic Art of World Famous Museums | 30 | 2 | 90 |
| 18-19/ Spring | College Physics IA | 64 | 4 | 93 |
| | Physical Education B | 32 | 1 | 86 |
| | Electrical and Electronic Technology | 56 | 3.5 | 92 |
| | Advanced Mathematics B | 80 | 5 | 92 |
| | Information retrieval and search engine | 16 | 1 | 85 |
| | Brand Positioning and Consumer Psychology | 32 | 2 | 84 |
| | Data Structures | 56 | 3.5 | 95 |
| | Wisdom in Military Science of Sun Tzu | 32 | 2 | 74 |
| | Fantastic Bionics | 28 | 2 | 83 |
| | Electrical and Electronic Experiments | 12 | 0.5 | 95 |
| | University Chinese | 32 | 2 | 92 |
| | English Reading & Writing | 32 | 2 | 86 |
| | Modern Chinese History | 48 | 3 | 90 |
| | English for International Communication | 32 | 2 | 87 |
| 18-19/ Summer | Startup Framework & Operations | 16 | 1 | 84 |
| | Cognition Learning of Computer Science Application | 1周 | 1 | 95 |

| Term | Course | Hours | Credit | Score |
|---|---|---|---|---|
| 19-20/Fall | Chinese Historical Geography | 28 | 2 | 97 |
| | Entrepreneurship 102: What can you do for your customer? | 16 | 1 | 92.5 |
| | Probability Theory and Mathematical Statistics | 40 | 2.5 | 100 |
| | Introduction to Mao Zedong Thought and Theoretical System of Socialism with Chinese Characteristic A | 32 | 2 | 81 |
| | Physical Education C | 32 | 1 | 83.4 |
| | Assembly Language Programming | 32 | 2 | 95 |
| | Digital Logic Design | 64 | 4 | 92.4 |
| | Design and Practice of software (A) | 32 | 2 | 94 |
| | Engineering Training (Electronic Engineering) | 2周 | 2 | 85.7 |
| | Physics Lab IA | 33 | 1.5 | 83 |
| 19-20/ Spring | Modern Algebra | 32 | 2 | 91.95 |
| | Computational Methods | 32 | 2 | 89 |
| | Computer Design and Practice | 56 | 3.5 | 95 |
| | Mathematical Logic | 32 | 2 | 98 |
| | Design and Analysis of Algorithms | 40 | 2.5 | 98 |
| | Principles of Computer Organization | 64 | 4 | 94.9 |
| | Introduction to Mao Zedong Thought and Theoretical System of Socialism with Chinese Characteristic B | 32 | 2 | 92.1 |
| | The Basic Principle of Marxism | 48 | 3 | 97 |
| | Practical course of Introduction to Mao Zedong Thought and Theoretical System of Socialism with Chinese Characteristic | 16 | 1 | 97.2 |
| | Speaking and Writing as a New Scientist | 32 | 2 | 98.5 |
| | Physical Education D | 32 | 1 | 84 |
| 19-20/ Summer | Introduction to Machine Learning | 32 | 2 | 88.5 |
| | Information Retrieval | 32 | 2 | 92 |
| 20-21/Fall | Deep Learning Architecture | 32 | 2 | 93.0 |
| | Computer Networks | 56 | 3.5 | 87 |
| | Artificial Intelligence | 40 | 2.5 | 94 |
| | Software Engineering | 32 | 2 | 92 |
| | Introduction to Information Security | 32 | 2 | 88 |
| | Operating System | 64 | 3 | 94 |

Director: Zhu Ning

Department of Academic Affairs
Harbin Institute of Technology,Shenzhen
Printing Date: May 07, 2021

哈尔滨工业大学(深圳) 学生证明专用章 (3)

哈爾濱工業大學(深圳)
HARBIN INSTITUTE OF TECHNOLOGY, SHENZHEN

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**Entrance Time:**August,2018

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**School:**School of Computer Science and Technology
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| Term | Course | Hours | Credit | Score | Term | Course | Hours | Credit | Score |
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